

Reference Dependence and Unobservable Poverty Traps

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Abstract

Decades of empirical research has found mixed results on the existence of poverty traps even when using long income and wealth panel data. I propose an explanation for the mixed evidence using behavioral theory. I model an intergenerational poverty trap where agents feel a loss if they bequest their progeny an amount less than their own initial endowment. This significantly slows and changes transitions into poverty. These dynamics are not accounted for by standard (existing) poverty trap models. I test two standard empirical identification strategies and in both cases I find the model rejects the null hypothesis a poverty trap exists despite its existence. This motivates further research into identification strategies and the theoretical dynamics that underlie poverty traps.

1 Introduction

Poverty traps describe a self-reinforcing mechanism where those below a certain wealth threshold face economic forces that infringe on their ability to generate wealth. This creates strong motivation for policy intervention and so identifying poverty traps is crucial. However, the evidence that poverty traps exist is mixed: while [Kraay and McKenzie \[2014\]](#) only find conclusive evidence for poverty traps in specific contexts, research by [Carter and Barrett \[2013\]](#) posit a greater prevalence of poverty traps but a dearth of sufficient data. It remains an outstanding question as to why even very rich empirical data are so rarely able to conclude the existence of poverty traps.

To help answer this question, I reassess an intergenerational poverty trap framework while importantly allowing for reference-dependent bequest motives to understand how behavioral interactions might change the existing poverty dynamics and therefore observation wealth and income data. My model borrows the framework from [Banerjee and Newman \[1993\]](#) where a mechanical poverty trap, separate from behavioral preferences, is generated within a credit market due to imperfect monitoring of loans. Agents begin their life with an initial wealth as a bequest from their parents. Those with low initial wealth are lent less, realize small incomes, and therefore bequest less to their progeny which transmits poverty. The key contribution of my model is that parents anchor their bequest expectations which I model using loss aversion [[Kahneman and Tversky, 1979](#)]. Loss aversion states that individuals feel a psychological loss much greater than an equivalent gain, and will change their behavior to avoid this felt loss. Agents anchor their expectation for how much they should bequest based on their own initial wealth and will feel a loss if they give less. The intuition is that parents want to give their children a life *at least as good* as their *own* initial conditions. Agents will sacrifice their consumption to increase their bequest to avoid this loss.

My model predicts a slower or entirely stagnant convergence to lower steady states than predicted by a model without loss aversion. Agents who normally give less than their own initial wealth increase their bequest, slowing downward convergence. If they can, an agent will increase

their bequest until it exactly matches their own initial wealth, avoiding the loss with the least possible increase, creating a *sticky* wealth level: dynasties at that initial wealth will bequest that initial wealth indefinitely, generating a new unstable steady state instead of converging towards a poorer stable steady state. Intervals of sticky wealth levels emerge close to steady states. These dynamics differ significantly from standard poverty trap models where all individuals below a threshold converge down to poverty.

I find that reference-dependence can make the poverty trap invisible to typical econometric empirical identification methods. I use Monte Carlo simulations on my model and, as a benchmark, my model without loss aversion to create a rich wealth time series panel. The most common and thought to be robust poverty trap identification method is the threshold test [Barrett and Carter, 2013], which identifies the unstable steady state where wealth growth paths bifurcate, the poorer branch drifting towards the poor stable state. First, I find that given the best wealth and income data, the threshold identifies a threshold but underestimates it. Second, I find that while the method identifies the threshold, it does not reliably identify those beneath the threshold converge to poverty; rather it indicates stagnant or slow *positive* wealth changes which is driven by the model's sticky wealth region and increased sensitivity to stochastic shocks. Despite the model's increase sensitivity to shocks, most dynasties initially below the threshold remain below the threshold. I confirm that without loss aversion, the empirical tests would otherwise accurately identify the scope and severity of the poverty trap.

The results of my model and empirical test offer new insight into poverty trap research. First, since even joint income and wealth data are insufficient to identify the trap, my model suggests supplementing these with consumption data might allow researchers to measure what was sacrificed to maintain the wealth level and therefore the economic headwind generated by the trap. Further, my model predicts a uniquely rational behaviors that could be identified with income and wealth data alone that might also help reveal the presence of a trap. Lastly, this paper's results indicate potential policy inefficiencies and ineffectiveness from failure to identify the trap or under-identification of its scope.

I begin in Section 2 by setting up the economic model and discuss the poverty trap dynamics. Then I discuss econometric implications in Section ?? and a discussion on future analysis in Section 4.

2 Model

2.1 Environment

Suppose a continuum of agents where the population is normalized to 1. Each agent enters the economy with some initial wealth in the form of a bequest. Time is discrete with an infinite horizon where the distribution of the initial wealth in time t is $G_t(w)$, and every period represents a generation.

I assume that agents are economically active for one period. At the beginning of the period, they enter the economy with their initial wealth, earn an income, then choose how much of their income to leave as a bequest to their child. By the end of the period all economic activity has occurred in equilibrium and all agents leave a bequest to their child, consume the rest, then they exit the economy. In the next period, after a child receives a bequest, they become an agent in the economy. Since every agent is replaced by only one child the population size is constant across periods. While this set-up uses discrete time, the remainder of this section discusses the optimal policy function for any given periods and the period notation is dropped until necessary.

Agents have homogeneous, risk-neutral preferences over bequests and consumption characterized by the utility function in Equation (1):

$$U(b, c; \rho_b, \rho_c) = \underbrace{\gamma \ln(b) + (1 - \gamma) \ln(c)}_{\text{Standard Cobb-Douglas Preferences}} + \underbrace{\eta [\gamma \cdot v(\ln(b) | \ln(\rho_b)) + (1 - \gamma) \cdot v(\ln(c) | \ln(\rho_c))]}_{\text{Loss-Averse Preferences}} \quad (1)$$

The first two terms represent standard Cobb-Douglas preferences over their bequest to their child, b (the “warm-glow” motive), and their consumption, c . The second two terms represent the loss-averse utility over their bequest and consumption. Loss aversion states that agents feel a “loss” greater than an equivalent “gain” relative to a reference point. Kőszegi and Rabin [2006] represent loss aversion using the gain-loss function $v(x|\rho)$ where utility input x has a reference point ρ . Agents gain utility $x - \rho$ when $x \geq \rho$ but lose utility $\lambda(x - \rho)$ when $x < \rho$, where $\lambda > 1$ is the loss-aversion coefficient. The value an agent places on loss-averse utility is $\eta > 0$. I define that agents are loss averse over their bequests and their consumption relative to ρ_b and ρ_c . The interpretation is that agents have an expectation for how much they *should* bequest to their child, and if they fail to meet that expectation they feel a loss (shame, guilt, compassion, etc.).

In this paper, I anchor expectations for bequests relative to an agent’s own initial wealth, $\rho_b = w$. I include loss aversion over consumption but always code it as a gain, $\rho_c = 0^+$, to preserve the standard Cobb-Douglas MRS when a bequest is coded as a gain and only deviate when a bequest is coded as a loss to isolate the effect of loss-averse bequests.

2.2 Production and Markets

Agents use their endowment to invest in a capital market, realize some return, then decide how much to consume and bequest. There is a single production technology that agents invest in and a lending market agents use to borrow capital for investment. Due to market imperfections, there is a probability that a borrower will renege on their loan and avoid repayment, which limits lending amounts.

All agents participate in the economy, using their initial wealth w as collateral to borrow k units of capital with global interest rate $r > 0$. Agents invest k and receive investment income $V(k) = R \min\{k, k^*\}$, $R > r$. Here, k^* is denoted as the “first-best” level of capital and an agent who borrows $k < k^*$ borrows their “second-best” level of capital.

After an agent realizes their investment income, they choose whether to repay the loan or renege. Agents repay their loan with interest kr in exchange for their collateral with interest, wr . The expected value of repayment is therefore: $k(R - r) + wr$.

Agents who renege on their loan forfeit their initial wealth and might successfully run away with their investment gains. If they renege there is some probability $\pi(k) = \min\{k, \bar{k}\}$ they are caught, where $k^* < \bar{k}$. The intuition is that the bigger the loan, the easier it is to catch. If caught, they face punishment $F = V(k)$ so the expected value of renegeing is $kR - k\pi \cdot kR$. Borrowers will repay if the expected value of repayment is greater than the expected value of renegeing: $k(R - r) + wr > kR - k^2\pi R$, which is the incentive-compatibility (IC) condition. The primary driver of renegeing is losing one’s collateral, so poorer agents have less to lose and therefore more likely to renege.

With perfect information, lenders know lend the amount where an agent is exactly indifferent between renegeing and repayment, and since poorer agents are more likely to renege, they are given smaller loans to meet that IC condition. Agents that want to borrow the first-best level of capital must have wealth $w \geq w^*$, which can be found by plugging k^* into the IC and rearranging: $w^* = k^* - \frac{\pi R k^{*2}}{r}$. And solving the IC for k , those below w^* receive capital $k(w) = \frac{r - \sqrt{r^2 - 4\pi R r w}}{2\pi R}$, their second-best loan. As the probability of catching those who renege increase, the lower the

necessary wealth to borrow k^* . This allows us to explicitly define an agent's income as a piecewise function around w^* :

$$I(w) = \begin{cases} k^*(R - r) + wr & \text{if } w \geq w^* \\ k(w)(R - r) + wr & \text{if } w < w^* \end{cases} \quad (2)$$

This income is split between bequests and consumption: $b + c = I(w)$, which is the agent's budget constraint. The amount of an agent's income spent on consumption or bequests is dependent on if they perceive their bequest as a gain or a loss.

2.3 Bequest Function

To understand the transmission of wealth from one generation to the next, we characterize the policy function for an agent as their bequest to their progeny. Agents are already differentiated by those who can borrow the first-best capital $w \geq w^*$ and those who borrow their second-best $w < w^*$. Now, agents in *both* wealth branches face another bifurcation: agents who feel a gain with their bequest $b \geq w$; and agents who feel a loss with their bequest $b < w$.

First, I consider when an agent codes their bequest as a gain, that is $b > w$. We take the MRS of Utility Equation (1). While the η term is still in the MRS equation, it cancels out in the numerator and denominator: $c^* = \frac{(1-\gamma)(1+\eta)}{\gamma(1+\eta)} b^* = \frac{1-\gamma}{\gamma} b^*$. Plugging c^* into the Income Equation (2), we obtain our optimal bequest when coded as a gain. I define this as the “baseline” bequest, $b_B(w)$, since it is simply the standard Cobb-Douglas propensity to bequeath: γ . Thus, $\forall w$ such that $b_B(w) \geq w$, then

$$b_B(w) = \begin{cases} \gamma[k^*(R - r) + wr] & \text{if } w \geq w^* \\ \gamma[k(w)(R - r) + wr] & \text{if } w < w^* \end{cases} \quad (3)$$

Next, I consider an agent who codes their bequest at a loss when their baseline bequest is less than their initial wealth, $b_B(w) < w$. In this case, $\lambda > 1$ and their propensity to bequest increases. The propensity to bequeath at a loss is: $\alpha = \frac{\gamma(1+\eta\lambda)}{1+\eta-\gamma\cdot\eta(1-\lambda)} > \gamma$.

Before we can define the full functional form of the “loss-averse” bequest function $b_L(w)$ we must first recognize that since then $\alpha I(w) > \gamma I(w)$ is an open interval where it is possible that $\alpha I(w) > w > \gamma I(w)$. Here, agents who might bequest $\alpha I(w) > w$ would bequest so much they no longer feel a loss and would like to decrease the bequest. But if they drop it to the baseline $\gamma I(w) < w$, they feel a loss again. The optimal bequest is therefore a corner solution where they give a bequest exactly equal to their own initial wealth $b(w) = w$. In other words, agents will increase their bequests until the minimum level they consider it a gain, which is their own initial wealth.

I define bequests that match initial wealth as “sticky” and these sticky wealth levels emerge as intervals throughout the wealth distribution. Intuitively, sticky intervals emerge near where the baseline bequest crosses the 45-degree line. At these wealth levels, an agent is already close to giving a bequest match, and it takes only a small increase to avoid the loss. We can see this in Figure (1), whereas the bequest function itself increases, the agents between the two crossing points of $\alpha I(w)$ and $\gamma I(w)$ agents bequest exactly their initial wealth, creating a sticky interval. Agents within the sticky region bequest $b(w) = w$, those below bequest $\alpha I(w)$ and those above $\gamma I(w)$.

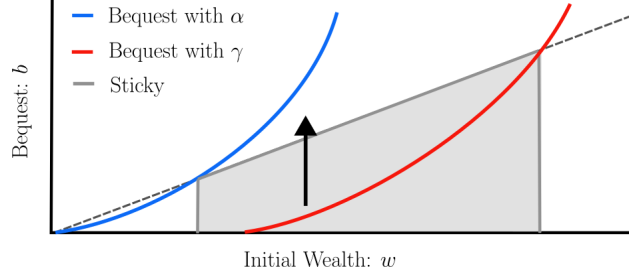


Figure 1: Bequest Changes and Sticky Interval

The full policy function, b , requires a careful stitching as sticky regions emerge in several places, causing new kinks to the baseline bequest function. I formulate the full bequest function in Proposition (1). If agents did not feel loss averse, the function below would collapse to the baseline case similar to [Banerjee and Newman \[1994\]](#). With loss aversion, the function complicates significantly despite the underlying mechanics of the economy being fairly simple.

Proposition 1 (Loss-Averse Bequests with Convex Capital Gains). *Given the previous model assumptions we define the bequest function as:*

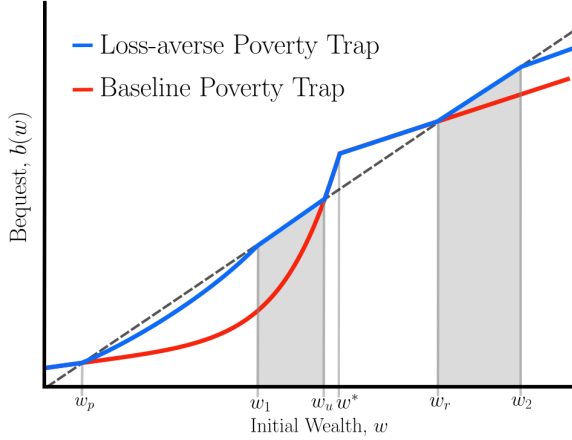
$$b(w) = \begin{cases} \alpha(k^*(R-r) + wr + T), & w_2 < w \leq \bar{w} \\ w, & w_r \leq w \leq w_2 \\ \gamma(k^*(R-r) + wr), & w^* \leq w \leq w_r \\ \gamma(k(w)(R-r) + wr), & w_u < w < w^* \\ w, & w_1 \leq w \leq w_u \\ \alpha(k(w)(R-r) + wr + T), & w_p \leq w < w_1 \end{cases}$$

The sticky intervals of wealth are defined as $S = \{[w_2, w_u], [w_r, w_3]\}$.

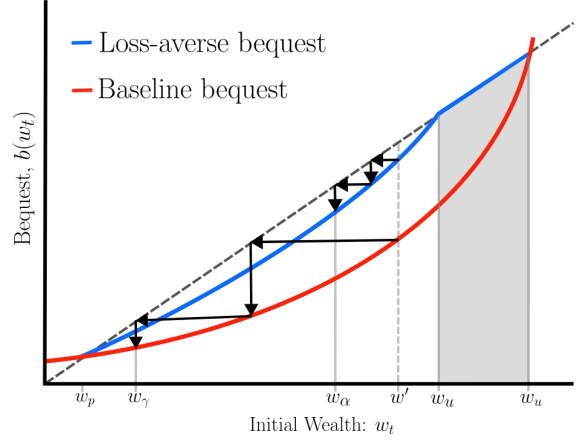
Proof. See appendix. □

While the above piecewise function looks complicated, ultimately the pieces simply describe when individuals are above or below w^* and when they do or do not feel a loss. For those $w < w^*$, those directly below might earn a second-best income but still earn enough to feel a gain in the bequest without changing in from the norm, $w_u < w < w^*$. Those just below the unstable steady state w_u *do* feel a loss, but are close enough to giving a bequest that can let them feel a gain, $b = w$, in some range $w_1 \leq w \leq w_u$. But for agents with wealth $w < w_1$, even increasing their bequest to $\alpha I(w)$, they cannot afford to match their bequest.

In Figure (2a) the red line is the “baseline” bequest function. The blue line is the “loss” bequest. In Figure 2a, the full bequest function is written out, where the blue line is the bequest when feeling a loss and the red the bequest when agents feel a gain.



(a) Poverty traps with loss-averse compared to baseline preferences



(b) Poverty trap transition differences below w_1 .

2.4 Poverty Trap

A poverty trap is defined as an economic mechanism that keeps agents below a certain threshold in poverty, unable to obtain a higher wealth or income. This threshold is denoted in this model as w_u . Since the bequest is the next generation's initial wealth, we can re-characterize the bequest function as a mapping from one period's wealth to the next: $b(w_t) = w_{t+1}$.

As seen in Figure 2b, there are notable difference between the *observed* poverty trap dynamics when agents have loss-averse versus standard preferences. An agent with an initial wealth $w \in [w_1, w_u]$ will keep their dynasty from falling into the poverty trap by maintaining their initial wealth every period, sacrificing present consumption to avoid a smaller bequest within this model. For agents with the same initial wealth without loss aversion, each generation receives a smaller bequest, realizes a smaller income, until they converge to abject poverty, w_p . For lineages in this model with an initial wealth $w' < w_1$, they still converge towards w_p but because each generation is increasing their bequest, convergence slows significantly as can be seen by the arrows in the diagram in Figure 2b which indicate the transitions of two periods, w_α . In the baseline model, convergence is quick: after two periods, that dynasty has initial wealth w_γ which has nearly converged to w_p whereas the other dynasty with loss aversion has barely drifted from their initial wealth.

Importantly, since loss aversion significantly reduces drifts in intergenerational wealth, changes in wealth levels are more sensitive to stochastic shocks. For all agents within the poverty trap, intergenerational wealth change might even be *positive* for a large enough positive shock, especially in the sticky interval which could even lead to escapes from the trap. While loss-averse bequests do weaken the long-run effects of the poverty trap, it is important to note it is at the cost of present consumption every period; we are weighing a large loss in consumption every period relative to that agent's preferred consumption in the baseline case, and evaluating which scenario is preferred is a philosophical question.

3 Econometric Implications

3.1 Benchmark Empirical Test: Threshold Test

The slowed and sticky intergenerational wealth levels might provide an explanation as to why even abundant wealth and income data might fail to identify a poverty trap when one exists. The most common and robust empirical poverty trap test is the threshold test [Jalan and Ravallion, 2004, Carter and Barrett, 2013, Antman and McKenzie, 2007]. This identifies a wealth level where the paths of average change in wealth bifurcate, one devolving into deeper poverty. We estimate two lines of best fit for data either side of the threshold, revealing their convergence paths and long-run states.

To test the efficacy of the threshold test on my bequest model, I design the empirical estimation model in Equation (4) based on the framework developed by Hansen [2000]. The true threshold is $c = w_u$ and the estimated threshold is $\hat{c}$. The line of best fit for wealth data greater than $\hat{c}$ is $g_h(w) = \beta_0 + \beta_1 w$ and lesser is $g_l(w) = (\beta_0 + \beta_2) + (\beta_1 + \beta_3)w$. We can reject the null hypothesis that a poverty trap *does not exist* if a threshold is identified where the trend lines show divergent trends, like that predicted by our S-shaped curve around w_u .

$$(T1) \quad \Delta w_{i,t+1} = \beta_0 + \beta_1 w_{i,t} + \beta_2 \mathbf{1}\{w_{i,t} < c\} + \beta_3 (w_{i,t} \cdot \mathbf{1}\{w_{i,t} < c\}) + \varepsilon_{i,t}. \quad (4)$$

I produce an ideal wealth and income panel using Monte Carlo simulations over 5 periods and 500,000 agents, significantly richer than even the best panel data previously used [Barrett and Carter, 2013]. I identify the threshold $\hat{c}$ using a grid search over a fine mesh of different c values and pick the $\hat{c}$ that minimizes the sum of squared errors for $\min_{\beta} ||Y - X(c)\beta||^2$.

The simulation results from Table 1 indicate that when agents have loss-averse bequests, the threshold test fails to reject the null hypothesis whereas the test can reject the null when agents are not loss averse. First, while the test correctly identifies a threshold, it is biased downward towards the edge of the sticky interval of wealth, $\hat{c} = w_1 = 0.842 < 0.848 = w_u$. Even if a poverty trap was identified, its scope could be under-estimated based on the *behavioral* threshold rather than the true. Second, due to the sticky wealth, estimated changes in wealth are close to zero and indeed are slightly positive around the threshold.

The overall flattening of transitions around the threshold is reflected by the estimated average change in wealth either side: $\hat{g}_l(\hat{c}) = 0.002$ and $\hat{g}_h(\hat{c}) = 0.093$. Instead of the threshold identifying immediate changes towards a poor and rich basin, like in the baseline where $\hat{g}_l(\hat{c}) = -0.120$ and $\hat{g}_h(\hat{c}) = 0.212$, transitions are predicted to be very slow but ultimately positive. A researcher would therefore conclude that around this threshold, poverty might be slow to escape but ultimately escapable: the threshold identifies a shift in acceleration rather than a bifurcation in paths.

Table 1: T1: Conventional threshold regression for downward convergence

Model	$\hat{c}$	$\hat{\beta}_0$	$\hat{\beta}_1$	$\hat{\beta}_2$	$\hat{\beta}_3$	$\hat{g}_l(\hat{c})$	$\hat{g}_h(\hat{c})$
Loss-averse	0.842	0.238	-0.173	-0.240	0.178	0.002	0.093
Baseline	0.848	0.793	-0.685	-0.795	0.545	-0.120	0.212

Notes: All coefficients are statistically significant within the 99% significance level. ‘‘Slope below’’ is the local slope of $E[\Delta w | w]$ for $w < \hat{c}$. In the loss-averse run, the slope below $\hat{c}$ is near zero (sticky).

I conduct a robustness check in Appendix (??).

4 Insights for New Analysis

Consumption Data. Despite the recommendation from [Barrett and Carter \[2013\]](#) to use both wealth and income data, this model indicates even that data, however rich, might still be insufficient. My model predicts that by incorporating consumption data, a researcher would be able to identify a parent’s sacrifice in order to maintain the anchored bequest. By measuring the distance between the expected consumption level and an actual income level given some income shock, this would allow researchers to measure headwinds, if that headwind is especially powerful, and if the headwind dissipates after a threshold. Further research should investigate the potential use of life-time income shocks (or, in another model, temporary income shocks) and the parental response in child investment. Other models, like that introduced by [Bolt et al. \[\(forthcoming\)\]](#), investigate many different types of parental investments and could be extended to investigate how loss-averse parents might be to those variety of investments, and how present and long-run consumption decisions change to match different expectations.

Uniquely Rational behavior. This model explores a behavior uniquely rational within both a poverty trap and under loss aversion that could help researchers identify the existence and scope of a poverty trap. Two agents shocked to an income level between their expected income levels will have different propensities potentially due to their initial wealth anchor. Without a poverty trap and with loss aversion, individuals positively shocked would not decrease their bequest propensity as this model predicts. Within a poverty trap and without loss aversion, negative and positive shocks would not change propensities. This creates a quasi-experimental method whereby aggregating many observed income-averaging shocks we could control for preference heterogeneity and potentially more reliably identify a possible trap.

Policy. This model also indicates severe policy implications. If a social welfare planner fails to identify a poverty trap, as this model shows is very possible, they might employ welfare-reducing policy. A typical distortionary tax on net income [[Mirrlees, 1971](#)] would negatively affect the IC condition (increasing w^*) in the capital market since small tax rates will have a larger negative influence on higher wealth levels, often those closer to w^* , increasing the scope of the poverty trap. One could formulate an optimal tax policy like in [[Piketty and Saez, 2013](#)], but this is unlikely to push many out of the poverty trap. The benefit of a one-period tax redistribution will be partially wiped out by the headwinds of the next period, causing long-term efficiency losses from continued ineffective rebate schemes. Even if a planner does identify the trap, the results from Table 1 indicate that the poverty trap threshold might be underestimated, $\hat{c} < w_u$. A “big-push” policy, a common poverty trap policy response that aims to give all agents below the threshold enough of an income supplement to rise just above it, thereby escaping the trap, might not be large enough. This would cause many generations to fall back into the trap by way of tiny negative shocks or fail to enter into upward mobility as the planner might expect, resulting in an ineffective policy with continued long-run needs for taxation and additional safety nets. Further research into poverty trap policy responses might benefit from the insights of this model.

5 References

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